

Camera Pose Estimation by Alignment from a Single Mountain Image

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Abstract. We present an alignment-based method for recovering camera position and orientation parameters from a single mountain image and digital elevation map. Camera pose estimation is achieved even without any initial position or orientation parameter estimate and requires only that the camera height above ground be known beforehand. It is also robust to partial occlusion. Using mountain peaks as features, image-model feature point alignment is hypothesized, producing a camera pose in the process. Each hypothesis is verified using mountain skyline-based geometric constraints. Combinatorial explosion is avoided using a strategy discussed in this paper. Probabilistic hypothesis generation is employed to guide the search process. Experiments involving synthetic and real images show that position accuracy compares favorably with existing algorithms.

1 Introduction

This paper addresses the problem of determining the position and orientation of a camera from an image of a mountain scene given the digital elevation map (DEM) of the surrounding terrain. This problem is of interest in vision-based navigation of robots in environments where Global Positioning System (GPS) signals are unreliable (e.g. tank navigation in the presence of enemy electronic countermeasures [19]) or unavailable (e.g. lunar or planetary rover navigation [4]). Vision-based position estimation algorithms require different types of prior position-dependent information. Several algorithms that compute position have been proposed using an initial position estimate [6] [1], accurate elevation value at viewpoint [17], etc. Most require multiple images, usually stitched together to form a panoramic view of the environment (e.g. [4],[16]).

We describe, in this paper, a camera pose estimation method that can compute position and orientation using a single mountain image without any estimate on camera location, elevation, and orientation. Although not required, knowledge that a certain mountain is visible in the image may be exploited to reduce computation time. It can also tolerate partial occlusion.

The novelty of this work lies in the application of the alignment paradigm to robot positioning and in the use of constraints appropriate to the domain. It is fundamentally different from competing methods in that it is capable of computing camera pose from a *single* mountain image and with minimum hardware requirements. Only a simple calibrated camera is needed. Other methods rely on multiple images which often need a more elaborate setup to ensure that the images are properly obtained (e.g. camera is kept horizontal or at a fixed tilt angle as it rotates about the vertical to obtain the panoramic image).

2 Pose Estimation Using Alignment

Our technique is based on the alignment principle which consists of a two-step hypothesize & verify process [10]. In the first stage, image plane alignment of a group of model (DEM) features with a group of image features is hypothesized. If it exists, the transformation or pose that maps the model features onto their corresponding image features is then computed. In the second stage, the transformation is applied to the model features so they can be compared directly with those of the image. Camera pose estimation then, is structured as a search for the correct pose that results in an optimal model feature-to-image feature match.

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Camera pose estimation by alignment is robust to partial occlusion since there is often a redundancy of image features. Spurious features introduced by noise and errors in the feature extraction process and by partial occlusion (e.g. trees, buildings, etc. in urban scenes) will increase the amount of search to be performed but will not affect the final result.

2.1 Mountain Peaks and Skyline as Features

A typical mountain image contains features that uniquely describe the mountain scene. These features, together with their locations on the image and their precise shapes, determine the viewpoint (position and orientation) of the camera that obtained the image.

Since we are not provided with any prior position-dependent information, we must rely only on position invariant features that can be obtained directly from both image and model in generating a transformation. Mountain peaks are appropriate local features since *mountain peaks are viewpoint-invariant* and can be extracted from image and model using some suitable operators. To reduce combinatorial complexity, the alignment technique dictates the use of a minimum number of image-model feature point correspondences in generating a transformation. Three model-image feature points correspondences are needed to compute the five camera pose parameters. The sixth parameter (the camera’s altitude) is not an independent variable since it is determined by the camera’s longitude, latitude, and height above ground. Three feature points correspondences, however, do not uniquely characterize a scene so that it is necessary to employ a global feature - the mountain skyline, whose precise shape completely determines viewpoint. The skyline will be the basis of some geometric constraints which we will use in reducing the search space and in verifying each generated hypothesis. We will also use the skyline in selecting the best among the candidate poses.

Our pose estimation process begins with the extraction of feature points corresponding to mountain peaks from both image and map model, as well as, the mountain skyline from the image. Image feature points corresponding to mountain peaks can be obtained from the skyline using curvature-based or derivative-based methods. The skyline can be extracted from the image using graph searching techniques such as A* [13], or dynamic programming on edges [2], etc. We used a MLP neural network-based skyline extractor that labels an edge pixel as skyline pixel when the pixels immediately above it are classified as “sky” pixels and pixels below it as “not sky” pixels. We then modeled image peaks as Gaussian in shape and developed a simple peak extraction procedure that searches for the feature point in the interval where the skyline’s second derivative values are negative. The area under the curve provides a measure of how “large” the peak is. Model feature points are obtained by comparing the elevation of each feature point candidate with the elevations of the points in a small circular area around it. We distinguish between *minor peaks* that are high elevation points in a small circular area around them and *major peaks* (the mountain’s most prominent peaks) which are the highest elevation points in a wider circular area. A typical mountain usually has one or two or sometimes three major peaks and several minor peaks visible in an image.

Model feature point extraction is done off-line and results are stored in two databases containing the absolute three dimensional coordinates of all major and minor peaks for the entire digital elevation map. The first database, which we call the Model Peak Database, contains only the DEM’s major peaks and the second (Model Feature Point Database) contains all major and minor peaks.

2.2 Pose Computation

Many analytical and iterative procedures have been proposed to compute for camera pose from three or more point correspondences and an initial pose estimate ([7], [9], [8], [11], etc). These procedures, however, cannot be used for our problem since no initial pose values are assumed.

We formulate pose computation, which involves the calculation of the position and orientation parameters that will best fit or align three model feature points with three image feature points, as a nonlinear least squares optimization problem. Since initial parameter values are unavailable, we employ a globally convergent nonlinear least squares optimization procedure. We model the imaging process using perspective projection, which describes image formation in an idealized pinhole camera. If greater accuracy is desired, a more elaborate camera model may be used provided some minor modifications are made to the procedure we describe below.

Let the camera position parameters $(x_{cam}, y_{cam}, z_{cam})$ and orientation parameters (pan, tilt, and swing angles, θ , ϕ , and ψ respectively) specify the viewpoint relative to a world reference frame. A world point

$\mathbf{p} = [x, y, z]^T$ is mapped into an image point $\mathbf{P} = [u, v]^T$ according to the following perspective transformation equations:

$$[\hat{x}, \hat{y}, \hat{z}]^T = \mathbf{R}(\mathbf{p} - \mathbf{t}) \quad (1)$$

$$\mathbf{P} = [P_u, P_v]^T = [u, v]^T = [f\hat{x}/\hat{z}, f\hat{y}/\hat{z}]^T \quad (2)$$

where $\mathbf{t} = [x_{cam}, y_{cam}, z_{cam}]^T$ and $[\hat{x}, \hat{y}, \hat{z}]$ are the camera-centered coordinates of the point, and $\mathbf{R}$ is the product of three rotation matrices, one for each axis.

Given N model feature point-to-image feature point correspondences, $\mathbf{p}_i = [x_i, y_i, z_i]^T \leftrightarrow \mathbf{P}_i = [u_i, v_i]^T$, $i = 1, \dots, N$, we now compute for the viewpoint parameter vector $\boldsymbol{\omega} = [x_{cam}, y_{cam}, z_{cam}, \phi, \theta, \psi]$ that maps $\mathbf{p}_i$ onto $\mathbf{P}_i$ using nonlinear least squares optimization.

Let the aligning transformation in Eqn. (2) be $[P_u(\mathbf{p}_i; \boldsymbol{\omega}), P_v(\mathbf{p}_i; \boldsymbol{\omega})]$ for $i = 1, \dots, N$. Also, let the distances between the image feature points and the model feature point projections on the image plane for pose $\boldsymbol{\omega}$ be represented by the vector

$$\mathbf{E}(\boldsymbol{\omega}) = [P_u(p_1; \boldsymbol{\omega}) - u_1, \dots, P_u(p_N; \boldsymbol{\omega}) - u_N, P_v(p_1; \boldsymbol{\omega}) - v_1, \dots, P_v(p_N; \boldsymbol{\omega}) - v_N]^T$$

To simplify notation, let $r_j(\boldsymbol{\omega})$ be the j th element of $\mathbf{E}(\boldsymbol{\omega})$, $j = 1, \dots, 2N$. Let the Jacobian of $\mathbf{E}(\boldsymbol{\omega})$ be denoted by $\mathbf{J}(\boldsymbol{\omega})$. We compute this Jacobian using its forward finite difference approximation since it is difficult to write an analytic expression.

Solving for the camera pose parameters $\boldsymbol{\omega}$ is the same as solving the nonlinear least squares optimization problem

$$\underset{\boldsymbol{\omega}}{\text{minimize}} \left\{ \|\mathbf{E}(\boldsymbol{\omega})\| = \frac{1}{2} \mathbf{E}(\boldsymbol{\omega})^T \mathbf{E}(\boldsymbol{\omega}) = \frac{1}{2} \sum_{j=1}^{2N} r_j(\boldsymbol{\omega})^2 \right\}$$

It can be solved iteratively using the Levenberg-Marquardt algorithm [5]:

$$\boldsymbol{\omega}^{(n+1)} = \boldsymbol{\omega}^{(n)} - (\mathbf{J}(\boldsymbol{\omega}^{(n)})^T \mathbf{J}(\boldsymbol{\omega}^{(n)}) + \mu^{(n)} \mathbf{I})^{-1} \mathbf{J}(\boldsymbol{\omega}^{(n)}) \mathbf{E}(\boldsymbol{\omega}^{(n)})$$

where $\mu^{(n)}$ is a nonnegative scalar the calculation of which is described in detail in [14]. Some versions of this algorithm (e.g. [15]) have been proven to be globally convergent, i.e. convergence is achieved from almost any initial starting point.

The value of $\|\mathbf{E}(\boldsymbol{\omega})\|$ at convergence, called the residual, measures the degree of misalignment of the three points.

It has been shown that the solution to the three world point-to-image point alignment problem is not unique [7]. We circumvent this problem by putting a constraint on the camera elevation variable. Since the height of the camera above ground is known, the camera elevation is not an independent variable but one that is completely determined by its longitude and latitude. This constrains the solution to one consistent with the physical problem.

2.3 K-Nearest Feature Point Search

Three model and image point correspondences are needed to compute the pose using the optimization procedure just described. For n model feature points and m image feature points, the number of hypotheses is $\mathcal{O}(n^3 m^3)$. This value exceeds 10^9 hypotheses even for a small (600 cell $\times$ 600 cell) digital elevation map.

An image typically captures one prominent mountain peak (i.e. major peak) together with other feature points (minor and major peaks). Physically, these feature points are generally close together in three dimensions. We make use of this spatial proximity property of feature points to formulate our *K-Nearest Feature Point Search Strategy*.

A mountain image must contain at least one major peak visible in the image. The model feature point corresponding to this major peak can then be used as basis for measuring proximity of the other model feature points. Let us call this model feature point the *model pivot point*. We then reduce the search effort by *imposing that one of the elements of the model feature point triple be a model pivot point and that the two other elements be feature points that are spatially close to the model pivot point*. We call the set of feature

points that are spatially proximate to the model pivot peak, the k -nearest feature point neighbors of the pivot peak, where k is a number much smaller than the total number of model feature points m .

Thus, instead of generating all the possible hypotheses, we reduce the search space by imposing the following constraints on the elements of the model feature point triple:

1. let the first model feature point in the triple be from the Model Peak Database (let us call this the model pivot point p_i),
2. let the next two feature points, p_j and p_k , in the triple come from the set of k -nearest feature point neighbors of the model pivot point. This set may also include major peaks and is stored in the Model Feature Point Database.

With this strategy, the number of hypotheses is reduced to n^3rs^2 , where r and s assume values much smaller than m . Our experiments show that reduction in search space size is about five to six orders of magnitude. The search for the consistent hypotheses is always performed on the search subspace where they are to be found.

The search strategy above accommodates prior information in the form of knowledge about the mountain present in the image. If it is known beforehand that a particular mountain is visible in the image, we can reduce the set of model pivot peaks for consideration to include only the model peaks for that mountain. Search space size reduction of one more order of magnitude is possible even for the small DEM used in our experiments.

2.4 Probabilistic Hypothesis Generation

The search for consistent hypotheses can be made more efficient by optimizing the order in which hypotheses are generated. Statistical information about the simultaneous visibility of peaks can guide the search order so that the most-likely hypotheses in the set of k -nearest feature point triples are generated first. The simultaneous visibility statistical distribution is obtained from a large set of images taken during a learning phase. In this learning process, images are taken at regular intervals in the general area where the model pivot point p_i is visible and these images are then examined for visibility of the feature points found in the Model Feature Point Database. We can therefore compute for the prior probabilities $P(p_i)$ for the model pivot peaks and the conditional model feature point probabilities $P(p_j|p_i)$.

From a given set of k -nearest feature point triples, the probability of observing the triple (p_i, p_j, p_k) is

$$P(p_j, p_k|p_i)P(p_i) = P(p_j|p_i)P(p_k|p_i)P(p_i). \quad (3)$$

Thus, instead of generating hypotheses from this set in random order, we can arrive at the consistent hypotheses faster if the generation of triples are ordered most-likely first following Eqn. (3).

2.5 Inter-Feature Point Visibility Constraint

Another geometric constraint involving two image feature points and their corresponding model feature points can further reduce the size of the remaining search space. It is based on the observation that if two image points are “visible to each other” (i.e. the line segment connecting these two image points is always above the skyline) then the world points that gave rise to them must also be “visible to each other” Thus, if this constraint holds for a pair of image feature points and their corresponding model points are blocked by intervening matter, then the hypothesis is inconsistent. We call this constraint the *Inter-Feature Point Visibility Constraint*. It is applied on the pairings of elements of the image feature point triple that satisfy the inter-image feature point visibility precondition. While the K-Nearest Feature Point Search Strategy imposes constraints on the elements of the model feature point triple, the Inter-Feature Point Visibility Constraint is a binary constraint applied on the elements of the image feature point triple.

2.6 Hypothesis Verification

Each pose computed by the globally convergent nonlinear least squares optimization procedure is verified using the mountain skyline. Although the computed pose may be used to generate a synthetic skyline from the model so a comparison between synthetic and real skylines could be made, synthetic skyline generation

is computationally expensive. Hypothesis verification is made more efficient through the use of a geometric constraint that eliminates poses that cannot possibly result in skyline matching before performing the synthetic skyline generation step.

Since the real skyline is the occluding contour of the terrain, all model points must project on the real skyline itself or below it. If one model point projects above the skyline, the pose should be eliminated. Further savings in computation can be achieved by projecting only the model feature points stored in the Model Feature Point Database. This constraint (which we call the “*Skyline is the Limit Constraint*”) remains valid even in the presence of occlusion.

Synthetic skylines are then generated for those poses that survive the constraint. The mean squared errors between synthetic and real skylines are then computed and used to select the best among a set of pose candidates.

3 Experiments

We conducted experiments with synthetic and real mountain images in order to experimentally validate the method and measure its performance. A mountain called Hieizan, near Kyoto City, was chosen for our experiments. Hieizan is a well-defined mountain having two major peaks. Twenty nine images of the mountain were obtained from 11 different locations using an off-the-shelf portable video camera.

The Model Peak Database and Model Feature Point Database were generated from a 600 cell $\times$ 600 cell DEM having a grid size of 50 meters and elevation resolution of 0.1 meter. These databases contained 34 major peaks and 213 major & minor peaks respectively. The number of image feature points extracted ranged from 3 to 15 but only the top six were considered in order to reduce the number of hypotheses. Selection of these best six was done automatically by the peak extraction procedure. The least squares residual cut-off was set to 2.0 pixels. A 15-nearest feature point search strategy was employed. The value $k = 15$ is not an optimized value. For comparison, synthetic images corresponding to each of the real images (i.e. having approximately the same pose) were generated and also processed. Position errors and processing time on a 170 MHz Sun Ultra Workstation are given in the tables below:

Position Errors (meters)			
	Min	Mean	Max
Synthetic Images	10	127	491
Real Images	86	393	1013

Processing Time (min:sec)			
Prior Info	Min	Mean	Max
w/ Prob Hyp Gen	0:5.5	0:30.7	2:45
w/o Prob Hyp Gen	0:35	4:28	13:44
none	14:30	96:10	295:55

Position errors for real images are significantly larger than for synthetic images since no attempt was made to model lens distortion. For comparison, other authors have reported position uncertainties of 95 meters [4] and 71,700 square meters [18] for a DEM grid size of 30 meters. In our experiments, however, the DEM we used has a grid size of 50 meters so an exact comparison of position accuracy is difficult to make.

For prior information, we used the fact that Hieizan is visible in the image so that only the two most prominent peaks of the mountain were considered as pivot peaks. The effect of this information on processing time is substantial and it does not affect position accuracy at all. Probabilistic hypothesis generation reduced the computation time enough to make it useful for some applications.

Processing rate is about 25.6 hypotheses/sec. The percentage of hypotheses rejected by the Inter-Feature Point Visibility Constraint is highly variable, depending mainly on viewpoint and presence of occlusion. This constraint is more effective when one of the mountains present in the scene is known. The results are summarized below:

Hypotheses Rejected by Visibility Constraint			
Prior Info	Min	Mean	Max
mountain known	7.0 %	20.9 %	41.9 %
none	2.6 %	5.2 %	15.7 %

The method also recovers the camera orientation parameters which may be used to visually confirm whether processing proceeded properly. In Fig. 1, the synthetic skyline (dark curve) corresponding to the best pose is superimposed on the image. This image, which contains partial occlusion, was taken with the camera deliberately rotated about its optical axis. Skyline alignment was achieved in spite of partial occlusion and camera rotation since three feature points were correctly extracted from the image.

4 Summary and Conclusion

In this paper, we described a method for recovering the camera position and orientation parameters from a single mountain image and digital elevation map (DEM) without the need for an initial pose estimate. It is based on the alignment paradigm. Using mountain peaks and the mountain skyline as features, image plane alignment of three image feature points with three model (i.e. DEM) feature points is first hypothesized. The pose for the hypothesis is then computed using globally convergent nonlinear least squares optimization and verified using skyline-based geometric constraints. We developed a search strategy that avoids combinatorial explosion. Probabilistic hypothesis generation orders the hypotheses most-likely first so that consistent hypotheses are obtained early. The method can successfully cope with partial occlusion in the image. Experiments involving real and synthetic images validated the method and gave encouraging results.

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Fig. 1.: Camera Orientation Result for an Image with Partial Occlusion and Rotation